

Tianzhe Chu

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Google Scholar · GitHub

RESEARCH HIGHLIGHTS

Continued Pretraining on Newly Published Books

Injects knowledge from newly published books into LLMs via continued pretraining with an instruction-following mix-in, preserving chat ability. Powers “Ask AI” on the book website and HKU DATA8014.
[website] [models]

Sycophancy in LLM-as-a-Judge for Competition Math

Benchmarks 14 LLM judges against humans on USAMO 2025 proofs, exposing consistent sycophantic overcrediting of plausible-but-flawed solutions and hallucinated logical bridges.
[website]

SFT Memorizes, RL Generalizes

Under equal training compute, RL with verified rewards generalizes to unseen textual and visual variants while SFT memorizes the training distribution, though SFT remains a necessary format-stabilization step before RL. Further shows that scaling up verification steps during RL training boosts its generalization.
[paper] [website] [code]

EDUCATION

The University of Hong Kong	Hong Kong SAR
<i>Ph.D. candidate at IDS advised by Prof. Yi Ma</i>	<i>Sep 2024 – Present</i>
ShanghaiTech University	Shanghai, China
<i>B.Eng in Computer Science and Technology (Honor)</i>	<i>Sep 2020 – Jun 2024</i>
University of California, Berkeley	Berkeley, CA, US
<i>Visiting Undergraduate, EECS & BAIR</i>	<i>Aug 2022 – May 2023</i>

PUBLICATIONS

(* denotes equal contribution)

SFT Memorizes, RL Generalizes: A Comprehensive Study of Foundation Model Post-training
Tianzhe Chu*, Yuexiang Zhai*, Jihan Yang, Shengbang Tong, Saining Xie, Dale Schuurmans, Quoc V. Le, Sergey Levine, Yi Ma
ICML 2025 [paper] [website] [code]

Pointer-CAD: Unifying B-Rep and Command Sequences via Pointer-based Edges & Faces Selection
Dacheng Qi, Chenyu Wang, Jingwei Xu, Tianzhe Chu, Zibo Zhao, Wen Liu, Wenrui Ding, Yi Ma, Shenghua Gao
CVPR 2026 [paper]

Seeing from Another Perspective: Evaluating Multi-View Understanding in MLLMs
Chun-Hsiao Yeh*, Chenyu Wang*, Shengbang Tong, Ta-Ying Cheng, Ruoyu Wang, Tianzhe Chu, Yuexiang Zhai, Yubei Chen, Shenghua Gao, Yi Ma
AAAI 2026 [paper] [website] [code]

White-Box Transformers via Sparse Rate Reduction: Compression Is All There Is?

Yaodong Yu, Sam Buchanan, Druv Pai, **Tianzhe Chu**, Ziyang Wu, Shengbang Tong, Hao Bai, Yuexiang Zhai, Benjamin D. Haeffele, Yi Ma
JMLR [[paper](#)]

Image Clustering via the Principle of Rate Reduction in the Age of Pretrained Models

Tianzhe Chu*, Shengbang Tong*, Tianjiao Ding*, Xili Dai, Benjamin D. Haeffele, René Vidal, Yi Ma
ICLR 2024 [[paper](#)] [[code](#)]

Emergence of Segmentation with Minimalistic White-Box Transformers

Yaodong Yu*, **Tianzhe Chu***, Shengbang Tong, Ziyang Wu, Druv Pai, Sam Buchanan, Yi Ma
CPAL 2024 (Oral), **NeurIPS 2023 XAI Workshop (Oral)** [[paper](#)] [[website](#)] [[code](#)]

White-Box Transformers via Sparse Rate Reduction

Yaodong Yu, Sam Buchanan, Druv Pai, **Tianzhe Chu**, Ziyang Wu, Shengbang Tong, Benjamin D. Haeffele, Yi Ma
NeurIPS 2023 [[paper](#)] [[website](#)] [[code](#)]

GenDexHand: Generative Simulation for Dexterous Hands

Feng Chen*, Zhuxiu Xu*, **Tianzhe Chu**, Xunzhe Zhou, Li Sun, Zewen Wu, Shenghua Gao, Zhongyu Li, Yanchao Yang, Yi Ma
Preprint [[paper](#)]

GAUSS Eval: Human-LLM Judge Consistency Analysis

Tianzhe Chu*, Jiaxin Zhang*, Zhenyu Liao, Qiuyu Ren, Tahsin Saffat, Zitong Yang, Yi Ma, Yue Zhang*
Blog Post [[website](#)]

OPEN SOURCE

Deep Representation Learning Book — Core Contributor [[code](#)]

Open-source textbook on the principles of deep representation learning.

CRATE — Core Contributor [[code](#)]

White-box transformer architecture derived from sparse rate reduction.

SELECTED AWARDS

Outstanding Graduate	Shanghai, China
<i>ShanghaiTech University</i>	<i>Jul 2024</i>
Merit Student	Shanghai, China
<i>ShanghaiTech University</i>	<i>Dec 2023</i>

INVITED TALKS

- Localize the Credits of Reasoning in Roaring 2025: LLM Reasoning Paper Potpourri @ ALICE, Aug. 2025
- SFT Memorizes, RL Generalizes @ BAAI, Jun. 2025
- SFT Memorizes, RL Generalizes @ Cathy Wu's lab, MIT, Apr. 2025
- White-Box Transformers @ SVIP Lab, ShanghaiTech, Sep. 2023
- Emergence of Segmentation in White-Box Transformers @ CPAL 2024, HKU, Jan. 2024

SERVICE

Head Teaching Assistant — HKU DATA 8014

Reviewer — ICLR 2025/2026, ICML 2025, NeurIPS 2025, CVPR 2026